Superstore Sales – Data Dictionary

Power BI project: clean_.pbix

1. Overview

This project analyses sales transactions of a US retail store (the "Superstore Sales Dataset"). Each record is one **product line inside a customer order**. The data was imported from a CSV file, cleaned in Power Query, and enriched with calculated columns, three dimension tables and 25 DAX measures.

Item	Value
Source	Superstore Sales Dataset.csv (18 raw columns)
Grain of the fact table	One row = one product on one order
Rows (fact table)	9,800
Orders	4,922
Customers	793
Products (Product ID)	1,861
Geography	49 states, 529 cities, 627 postal codes, 4 regions
Order dates	03 Jan 2015 – 30 Dec 2018
Last ship date	05 Jan 2019
Total sales	2,261,536.78 (USD)
Tool	Power BI Desktop (Power Query + DAX)

Business questions the data supports: sales trend over time, performance by category / sub-category / region / state, customer behaviour (repeat vs one-time customers, top customers), order size distribution, and shipping performance.

Limitation: the only numeric business value is `Sales`. The dataset has no Quantity, Discount or Profit, so profitability analysis is not possible.

2. Model Tables at a Glance

Table	Type	Rows	Purpose
Superstore Sales Dataset	Fact (transactions)	9,800	Order-line sales with customer, product, location and shipping details
Dim_Customer	Dimension (DAX calculated table)	793	One row per customer
Dim_Product	Dimension (DAX calculated table)	1,893	One row per product (Product ID and Product Name) with category hierarchy
Dim_Date	Dimension (DAX calculated table)	1,464	Daily calendar from first order date to last ship date
Sales measures	Measure holder (empty table)	–	Sales KPIs
Customer Analysis measures	Measure holder (empty table)	–	Customer KPIs
shipping	Measure holder (empty table)	–	Shipping KPIs
LocalDateTable_… and DateTableTemplate_…	Auto-generated by Power BI	–	Hidden auto date/time tables (not business tables)

3. Fact Table: Superstore Sales Dataset

22 columns: 16 loaded from the CSV and 6 calculated columns.

#	Column	Data type	Description	Example	Notes
1	Order ID	Text	Order number. Format: prefix (CA or US), order year, number	CA-2017-106075	4,922 distinct. Repeats on every product line of the order
2	Order Date	Date	Date the order was placed	2017-09-18	Converted from text dd/MM/yyyy
3	Ship Date	Date	Date the order was shipped	2017-09-23	Converted from text dd/MM/yyyy. Never earlier than Order Date
4	Ship Mode	Text	Shipping service level	Standard Class	4 values: Standard Class, Second Class, First Class, Same Day
5	Customer ID	Text	Unique customer code	HM-14980	793 distinct. Key to Dim_Customer
6	Customer Name	Text	Customer full name	Henry MacAllister	One name per Customer ID
7	Segment	Text	Customer segment	Consumer	3 values: Consumer, Corporate, Home Office
8	City	Text	Delivery city	New York City	529 distinct
9	State	Text	Delivery US state	New York	49 distinct
10	Postal Code	Text	Delivery ZIP code	10009	Stored as text. 4-digit codes padded with a leading 0. Missing values set to "unknown"
11	Region	Text	Sales region	East	4 values: Central, East, South, West
12	Product ID	Text	Unique product code. Format: category – sub-category – number	OFF-BI-10004654	1,861 distinct. Not a relationship key (see Product Key)
13	Category	Text	Top-level product category	Office Supplies	3 values: Furniture, Office Supplies, Technology
14	Sub-Category	Text	Product sub-category	Binders	17 values. Each belongs to exactly one Category

#	Column	Data type	Description	Example	Notes
15	Product Name	Text	Product description	Avery Binding System Hidden Tab Executive Style Index Sets	1,849 distinct names
16	Sales	Decimal	Sales amount of this line in USD	4.616	Min 0.44, average 230.77, max 22,638.48. No zero or negative values
17	Shipping Days	Whole number (calculated)	Days between order and shipping	5	Range 0–7
18	Shipping Speed	Text (calculated)	Shipping speed class based on Shipping Days	Slow (5+ days)	Fast (0-2 days), Normal (3-4 days), Slow (5+ days)
19	Order Total	Decimal (calculated)	Total sales of the whole order, repeated on every line of that order	4.616	Order-level value on a line-level table. Do not SUM it. Hidden in report view
20	Order Range	Text (calculated)	Order size band based on Order Total	1) Under 100	1) Under 100, 2) 100 - 500, 3) 500 - 1000, 4) Over 1000
21	Year	Whole number (calculated)	Year of the order	2017	2015–2018
22	Product Key	Text (calculated)	Unique product key made of Product ID and Product Name	OFF-BI-10004654 | Avery Binding System Hidden Tab Executive Style Index Sets	1,893 distinct. Key to Dim_Product. Needed because some Product IDs have two names

Calculated column logic

Column	DAX logic
Shipping Days	`DATEDIFF([Order Date], [Ship Date], DAY)`
Shipping Speed	`SWITCH(TRUE(), [Shipping Days] <= 2, "Fast (0-2 days)", [Shipping Days] <= 4, "Normal (3-4 days)", "Slow (5+ days)")`
Order Total	`CALCULATE(SUM(Sales), ALLEXCEPT('Superstore Sales Dataset', [Order ID]))`

Column	DAX logic
Order Range	`SWITCH(TRUE(), [Order Total] < 100, "1) Under 100", [Order Total] < 500, "2) 100 - 500", [Order Total] < 1000, "3) 500 - 1000", "4) Over 1000")`
Year	`YEAR([Order Date])`
Product Key	[Product ID] & " | " & [Product Name]

4. Dimension Tables

Dim_Customer (793 rows, one per customer)

Column	Data type	Description
Customer ID	Text	Unique customer code (primary key)
Customer Name	Text	Customer full name
Segment	Text	Consumer, Corporate or Home Office

Built from the distinct Customer IDs of the fact table.

Dim_Product (1,893 rows, one per Product ID and name)

Column	Data type	Description
Product Key	Text	Product ID and Product Name joined together (primary key)
Product ID	Text	Product code (not a relationship key)
Product Name	Text	Product description
Category	Text	Furniture, Office Supplies or Technology
Sub-Category	Text	17 sub-categories

Built with SUMMARIZE from the fact table. Product Key is the unique key, so the 32 Product IDs that carry two names in the source appear as two rows and their sales are not merged.

Dim_Date (1,464 rows, one per day)

Column	Data type	Description
Date	Date	Calendar day, from the first Order Date (2015-01-03) to the last Ship Date (2019-01-05). Primary key
Year	Whole number	Calendar year
Month No	Whole number	Month number 1–12
Month	Text	Month name (January … December)
Quarter	Text	Q1 – Q4

5. DAX Measures (25)

Sales measures

Measure	Definition	Meaning
Total Sales	`SUM(Sales)`	Total sales revenue
Total Orders	`DISTINCTCOUNT(Order ID)`	Number of distinct orders
Total Products	`DISTINCTCOUNT(Product ID)`	Number of distinct products sold
Total Records	`COUNTROWS(fact)`	Number of order lines
Average Order Value	`[Total Sales] / [Total Orders]`	Average sales per order
Sales per Order Line	`[Total Sales] / COUNTROWS(fact)`	Average sales per order line
Average Items per Order	`COUNTROWS(fact) / [Total Orders]`	Average number of lines in an order

Measure	Definition	Meaning
Highest Order Value	`MAXX(VALUES(Order ID), [Total Sales])`	Sales of the biggest order in the current filter
Lowest Order Value	`MINX(VALUES(Order ID), [Total Sales])`	Sales of the smallest order in the current filter
Large Orders (Over 1000)	`COUNTROWS(FILTER(VALUES(Order ID), [Total Sales] > 1000))`	Number of orders above 1,000
Sales % of Total	`[Total Sales] / CALCULATE([Total Sales], ALLSELECTED(fact))`	Share of the sales in the current selection
Sales Last Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Sales of the same period in the previous year
YoY Growth %	`([Total Sales] - [Sales Last Year]) / [Sales Last Year]`	Year-over-year growth. Blank for 2015 (no previous year)
Tech Avg Sales per Record	Total Sales / Total Records, filtered to Category = Technology	Average sales per order line in Technology

Customer Analysis measures

Measure	Definition	Meaning
Total Customers	`DISTINCTCOUNT(Customer ID)`	Number of distinct customers
Repeat Customers	Customers with more than 1 distinct order	Returning customers
One-Time Customers	`[Total Customers] - [Repeat Customers]`	Customers with a single order
Repeat Customer %	`[Repeat Customers] / [Total Customers]`	Share of returning customers
Orders per Customer	`[Total Orders] / [Total Customers]`	Average number of orders per customer

Measure	Definition	Meaning
Average Sales per Customer	`[Total Sales] / [Total Customers]`	Average revenue per customer
Top 10 Customers Share	Sales of the top 10 customers / total sales	Revenue concentration in the top 10 customers
Top 50 Customers Sales	Sales of the top 50 customers	Revenue from the top 50 customers

shipping

Measure	Definition	Meaning
Avg Shipping Days	`AVERAGE(Shipping Days)`	Average days from order to shipping
Max Shipping Days	`MAX(Shipping Days)`	Slowest shipment
Min Shipping Days	`MIN(Shipping Days)`	Fastest shipment

6. Relationships

The model is a star schema. Each dimension filters the fact table through a one-to-many, single-direction relationship, with the dimension on the "one" side.

From (one)	To (many)	Status
Dim_Customer[Customer ID]	Superstore Sales Dataset[Customer ID]	Active
Dim_Product[Product Key]	Superstore Sales Dataset[Product Key]	Active
Dim_Date[Date]	Superstore Sales Dataset[Order Date]	Active

7. Data Preparation (Power Query)

1. Loaded the CSV (UTF-8, comma delimited, 18 columns) and promoted the first row to headers.
2. Set data types for all columns.

3. Removed `Row ID` and `Country`.
4. Converted `Postal Code` to text and added a leading 0 to 4-digit codes.
5. Replaced missing postal codes with "unknown".
6. Parsed `Order Date` and `Ship Date` from text (dd/MM/yyyy) to real dates.
7. Renamed and reordered the columns.

8. Data Quality Notes

- **Completeness:** no empty values in any column of the fact table.
- **Valid dates:** no order ships before it is ordered.
- **Duplicate row:** one line of order US-2015-150119 (product FUR-CH-10002965, sales 281.37) appears twice with identical values. It was kept on purpose because it may be a legitimate repeated line (for example, the same item entered twice on one order), so its sales are included in all totals.
- **Missing postal code:** Burlington, Vermont has no postal code in the source and is labelled "unknown".
- **Product names:** some Product IDs carry two different product names in the source. This is solved by Product Key (Product ID and Product Name), which keeps each name as its own row in Dim_Product.
- **Postal code and city:** a postal code can map to more than one city name, so Postal Code alone is not a safe geography key.
- **Order Total:** repeats on every line of an order. Summing it overcounts, so the column is hidden in report view; use `Total Sales` instead.
- **Partial calendar years:** Dim_Date starts on 3 Jan 2015 and ends on 5 Jan 2019, so 2015 and 2019 are incomplete years.